Т3.

$$p(x) = \begin{cases} \frac{e^{-\frac{x}{\theta}}}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \theta > 0$$

$n = 3$

θ: $\tilde{\theta}_1 = \bar{x}$

$\tilde{\theta}_3 = x_{(2)}$

$\xi \sim F(x)$: $F'(x) = p(x)$

$$M\xi = \int_{-\infty}^{+\infty} x\, dF(x) = \int_0^{+\infty} x \cdot \frac{e^{-\frac{x}{\theta}}}{\theta} dx =$$

$$= -\int_0^{+\infty} x e^{-\frac{x}{\theta}} d\left(-\frac{x}{\theta}\right) = -\int_0^{+\infty} x\, de^{-\frac{x}{\theta}} = -\underbrace{x e^{-\frac{x}{\theta}}\Big|_0^{+\infty}}_{0} + \int_0^{+\infty} e^{-\frac{x}{\theta}} dx =$$

$$= -\theta \cdot e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = \theta$$

$$M\xi^2 = \int_0^{+\infty} x^2 \frac{e^{-\frac{x}{\theta}}}{\theta} dx = \int_0^{+\infty} 2x e^{-\frac{x}{\theta}} dx = 2\theta \cdot \int_0^{+\infty} e^{-\frac{x}{\theta}} dx =$$

$$= 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = \theta^2$$

а) $M\tilde{\theta}_1 = \frac{1}{n}\sum_{i=1}^{n} M\xi = M\xi = \theta$ – несмещённая.

б) $M\tilde{\theta}_3 = ?$

$\tilde{\theta}_3 \sim \varphi(x)$; $\varphi(x) = \sum_{i=2}^{3} C_3^i F^i(x)(1-F(x))^{n-i} =$

$$= 3 \cdot F^2(x)(1-F(x)) + F^3(x) = 3F^2(x) - 2F^3(x)$$

$$\varphi(x) = 6F(x)p(x) - 6F^2(x)p(x) = \varphi'(x); \quad F(x) = \int p(x)dx = \cancel{\int \frac{e^{-\frac{x}{\theta}}}{\theta}}$$

$$\cancel{M\tilde{\theta}_3 = \int_0^{+\infty} x\varphi}$$

$$F(x) = \begin{cases} \int_0^x \frac{e^{-\frac{t}{\theta}}}{\theta} dt, & x>0 \\ 0, & x<0 \end{cases}$$

$$\int_0^x \frac{e^{-\frac{t}{\theta}}}{\theta} dt = -e^{-\frac{t}{\theta}}\Big|_0^x = -e^{-\frac{x}{\theta}} + 1 = 1 - e^{-\frac{x}{\theta}}$$

$$\varphi(x) = 6\cdot\left(1-e^{-\frac{x}{\theta}}\right)\cdot\frac{e^{-\frac{x}{\theta}}}{\theta} - 6\left(1-e^{-\frac{x}{\theta}}\right)^2\cdot\frac{e^{-\frac{x}{\theta}}}{\theta} =$$

$$= 6\left(1-e^{-\frac{x}{\theta}}\right)p(x)\cdot\left(e^{-\frac{x}{\theta}}\right) = 6\cdot\frac{e^{-\frac{2x}{\theta}}}{\theta} - 6\cdot\frac{e^{-\frac{3x}{\theta}}}{\theta}$$

$$M\tilde{\theta}_3 = \int_0^{+\infty} x\frac{6}{\theta}\left(e^{-\frac{2x}{\theta}} - e^{-\frac{3x}{\theta}}\right)dx =$$

$$= -3\int_0^{+\infty} xe^{-\frac{2x}{\theta}} d\left(\frac{2x}{\theta}\right) + 2\int_0^{+\infty} xe^{-\frac{3x}{\theta}} d\left(\frac{3x}{\theta}\right) =$$

$$= \underbrace{-3e^{-\frac{2x}{\theta}}x\Big|_0^{+\infty}}_{0} + \underbrace{2e^{-\frac{3x}{\theta}}x\Big|_0^{+\infty}}_{0} + 3\int_0^{+\infty} e^{-\frac{2x}{\theta}}dx - 2\int_0^{+\infty} e^{-\frac{3x}{\theta}}dx =$$

$$= -\frac{3\theta}{2}e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} + \frac{2\theta}{3}e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} = +\frac{3\theta}{2} - \frac{2}{3}\theta = \frac{5}{6}\theta$$ – смещённая

Поправим:

$$\tilde{\theta}_2 = \frac{6}{5}\tilde{\theta}_3 \Rightarrow M\tilde{\theta}_2 = \theta$$ – несмещённая.

$$= \left(\frac{1}{2}-\theta\right)\cdot\left(1-\frac{1}{2}+\theta\right)$$

б) $D\tilde{\theta}_1$ - ?

$\cancel{D\tilde{\theta}_1 = \frac{1}{n^2} D\tilde{\theta}_1 \cdot n = \frac{D\tilde{\theta}_1}{n} = \theta^2}$

$$D\tilde{\theta}_1 = D\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n^2}\, n D\xi = \frac{D\xi}{n} = \frac{\theta^2}{n} = \frac{\theta^2}{3}$$

$$M\tilde{\theta}_3^2 = \int_0^{+\infty} x^2 \cdot 6 \cdot \frac{1}{\theta}\left(e^{-\frac{2x}{\theta}} - e^{-\frac{3x}{\theta}}\right) dx =$$

$$= 3 \cdot \int_0^{+\infty} 2x e^{-\frac{2x}{\theta}} dx - 2\int_0^{+\infty} 2x e^{-\frac{3x}{\theta}} dx =$$

$$= -3\cdot\theta \int_0^{+\infty} x\, de^{-\frac{2x}{\theta}} + \frac{4}{3}\theta \int_0^{+\infty} x\, de^{-\frac{3x}{\theta}} =$$

$$= 3\theta \int_0^{+\infty} e^{-\frac{2x}{\theta}} dx - \frac{4}{3}\theta \int_0^{+\infty} e^{-\frac{3x}{\theta}} dx =$$

$$= -\frac{3}{2}\theta^2 \cdot e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} + \frac{4}{9}\theta^2 e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} =$$

$$= \frac{3}{2}\theta^2 - \frac{4}{9}\theta^2 = \frac{27-8}{18}\theta^2 = \frac{19}{18}\theta^2$$

$$D\tilde{\theta}_3 = M\tilde{\theta}_3^2 - (M\tilde{\theta}_3)^2 = \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{\theta^2}{36}\cdot(38-25) = \frac{13\theta^2}{36}$$

$D\tilde{\theta}_2 > D\tilde{\theta}_1 \Rightarrow \tilde{\theta}_1$ эффективнее оценивает параметр θ

$$D\tilde{\theta}_2 = \frac{36}{25}\cdot D\tilde{\theta}_3 = \frac{13}{25}\theta^2$$

$$\frac{13}{25} - \frac{1}{3} = \frac{39-25}{75} > 0 \Rightarrow \frac{13}{25} > \frac{1}{3}$$

c) 1) регулярность модели:

1.1) $p(x)$ - непрер. дифф. $\oplus$

1.2) $\underbrace{\frac{\partial}{\partial\theta}\int_{-\infty}^{+\infty} dF(x)}_{=0} = \underbrace{\int_{-\infty}^{+\infty} d\left(\frac{\partial}{\partial\theta}F(x)\right)}_{\int_{-\infty}^{+\infty}\frac{\partial p}{\partial\theta}dx}$; $\frac{\partial}{\partial\theta}\left(\frac{e^{-\frac{x}{\theta}}}{\theta}\right) = \left(\frac{\frac{x}{\theta^2}e^{-\frac{x}{\theta}}\theta - e^{-\frac{x}{\theta}}}{\theta^2}\right) = -\frac{e^{-\frac{x}{\theta}}(x-\theta)}{\theta^3}$

$\cancel{\int_0^{+\infty}\frac{e^{-\frac{x}{\theta}}(x-\theta)}{\theta^3}dx}$

$\cancel{\frac{1}{\theta^2}\cdot\theta + \frac{1}{\theta}e^{-\frac{x}{\theta}} = \frac{1}{\theta} - \frac{1}{\theta} = 0}$ - модель не является регулярной. $\Rightarrow$

$\cancel{\Rightarrow}$ с помощью нер-ва Крамера-Рао оценить нельзя

$$\int_0^{+\infty}\frac{e^{-\frac{x}{\theta}}(x-\theta)}{\theta^3}dx = \frac{1}{\theta^2}M\xi - \frac{1}{\theta^2}\int_0^{+\infty}e^{-\frac{x}{\theta}}dx =$$

$$= \frac{1}{\theta^2}\cdot\theta + \frac{1}{\theta}e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = \frac{1}{\theta} - \frac{1}{\theta} = 0$$

$\cancel{\Rightarrow}$ модель является регулярной.

1.3) Информация Фишера:

$$I(\theta) = \int\limits_{A(\theta)} \left(\frac{\partial \ln p(x,\theta)}{\partial \theta}\right)^2 p(x,\theta)\,dx$$

$$\ln p(x,\theta) = \ln\left(\frac{e^{-\frac{x}{\theta}}}{\theta}\right) = \ln\left(e^{-\frac{x}{\theta}}\right) - \ln(\theta) =$$

$$= -\frac{x}{\theta} - \ln\theta$$

$$\frac{\partial}{\partial\theta}\ln p(x,\theta) = \frac{x}{\theta^2} - \frac{1}{\theta}$$

$$I(\theta) = \int\limits_0^{+\infty}\left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2 \frac{e^{-\frac{x}{\theta}}}{\theta}\,dx =$$

$$= \int\limits_0^{+\infty}\frac{x^2}{\theta^5}e^{-\frac{x}{\theta}}dx - 2\int\limits_0^{+\infty}\frac{x}{\theta^4}e^{-\frac{x}{\theta}}dx + \int\limits_0^{+\infty}\frac{1}{\theta^3}e^{-\frac{x}{\theta}}dx =$$

$$= \frac{1}{\theta^4}M\xi^2 - 2\frac{1}{\theta^3}M\xi + \frac{1}{\theta^2}\cdot\left(-e^{-\frac{x}{\theta}}\right)\Big|_0^{+\infty} =$$

$$= \frac{1}{\theta^4}\cdot 2\theta^2 - \frac{2}{\theta^3}\cdot\theta + \frac{1}{\theta^2} = \frac{1}{\theta^2}$$ – непрер, положительна.

$\Rightarrow$ модель регулярна.

2) Проверяем регулярность оценок:

$D\tilde{\theta}_1 = \frac{\theta^2}{3}$ – огранич. на $\forall$ компакте из $\Theta = (0,+\infty)$

$D\tilde{\theta}_2 = \frac{\theta^2}{10}$ – ——— || ———

и т.к. модель регулярна, а оценки несмещ., то обе являются регулярными.

Тогда применимо нер-во Крамера-Рао:

1) $D(\tilde{\theta}_1) = \frac{\theta^2}{3} = \frac{(\theta^2)^2}{3 \cdot I(\theta)} = \frac{\theta^2}{3}$ $\Rightarrow$ эффективная

$D(\tilde{\theta}_2) = \frac{11\theta^2}{25} > \frac{\theta^2}{3}$

~~Т.к.~~

$\cancel{\rho(x) = \frac{1}{\theta} \cdot \{(-1,0) \cup (0,1)\} + \alpha\{0,2\}}$

Т4.

$p(x) = a\{[-1,0)\cup(0,1]\} + b\{0;2\}$

1) $\int_{-1}^{1} a\,dx + 2b = 1$

$2a + 2b = 1$

$a + b = \frac{1}{2}$

$b = \frac{1}{2} - a$; $a = \theta$ - параметр

$b \in (0, \frac{1}{2}) \Rightarrow \theta \in (0, \frac{1}{2}) = \Theta$

2) метод моментов:

$\alpha_1 = M\xi = \int_{-1}^{1} \theta x\,dx + 2\cdot(\frac{1}{2}-\theta) + 0\cdot(\frac{1}{2}-\theta) =$

$= \theta\cdot 0 + 1 - 2\theta = 1 - 2\theta$

$\tilde{\alpha}_1 = \bar{x} = \alpha_1 \Rightarrow 1 - 2\tilde{\theta}_1 = \bar{x}$

$\tilde{\theta}_1 = \frac{1-\bar{x}}{2}$ - оценка параметра

3) метод макс. правдоподобия.

$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta)$

Пусть m раз выпал 0 или 2.

$L(\theta) = (\frac{1}{2}-\theta)^m \cdot \theta^{n-m}$

$\ln L(\theta) = m\ln(\frac{1}{2}-\theta) + (n-m)\ln\theta$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \frac{-m}{\frac{1}{2}-\theta} + \frac{n-m}{\theta} = 0$$ — ищем стац. точки

$$+\theta m = (n-m)\left(\frac{1}{2}-\theta\right)$$

$$\theta m = \frac{n-m}{2} - \theta n + \theta m$$

$$\tilde{\theta} = \frac{n-m}{2n} = \frac{1}{2}(1-\nu),$$ где $\nu = \frac{m}{n}$ — частота появления орла

оценка максимума

$$\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} = \frac{-m}{\left(\frac{1}{2}-\theta\right)^2} - \frac{n-m}{\theta^2} < 0 \Rightarrow$$ точка максимума.

4) несмещённость

$$M\tilde{\theta}_1 = M\left(\frac{1-\frac{1}{n}\sum_{i=1}^{n} x_i}{2}\right) = \frac{1-M\xi}{2} = \left\{M\xi = 1-2\theta\right\} = \theta$$ несмещ.

$$M\tilde{\theta}_2 = \frac{1}{2} - \frac{1}{2}M\nu = \frac{1}{2} - \frac{1}{2}\underbrace{p}_{\frac{1}{2}-\theta}$$

$$M\nu = M\left(\frac{1}{n}\cdot\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\sum_{i=1}^{n} Mx_i = \frac{1}{n}\cdot\sum_{i=1}^{n}\left(\underbrace{\left(\frac{1}{2}-\theta\right)}_{2\cdot\left(\frac{1}{2}-\theta\right)\cdot 1} + \theta\cdot(-\theta)\right) = \frac{1}{n}\cdot 2\theta$$

$$M\tilde{\theta}_2 = \frac{1}{2} - \frac{1}{2}(1-2\theta) = \theta$$ — несмещ.

5) состоятельность:

$$D\tilde{\theta}_1 = D\left(\frac{1}{2} - \frac{1}{2n}\sum_{i=1}^{n} x_i\right) = \frac{1}{4n}D\xi$$

$$M\xi^2 = \int_{-1}^{1} \theta x^2 dx + 4\cdot\left(\frac{1}{2}-\theta\right) = \frac{2}{3}\theta + 2 - 4\theta = 2 - \frac{10}{3}\theta$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2 - \frac{10}{3}\theta - 1 + 4\theta - 4\theta^2 = 1 + \frac{2}{3}\theta - 4\theta^2$$

$$D\tilde{\theta}_1 = \frac{1+\frac{2}{3}\theta-4\theta^2}{4n} \xrightarrow[n\to\infty]{} 0$$ несмещ. $\Rightarrow$ по дост. усл. состоятельна

$D\tilde{\theta}_2 = D\left(\frac{1}{2} - \frac{1}{2}\nu\right) = \frac{1}{4} D\nu$

$D\nu = \frac{1}{n^2} \sum_{i=1}^{n} D(\eta)$

$x_i \Rightarrow \eta_i$ – независимые случ. вел.

$\eta = \begin{cases} 1, & x_i = \{0,2\} \\ 0, & x_i \neq \{0,2\} \end{cases}$, т.к. $p_0 + p_2$

$M\eta = \left(\frac{1}{2} - \theta\right) \cdot 2 = 1 - 2\theta$

$M\eta^2 = \left(\frac{1}{2} - \theta\right) \cdot 2 = 1 - 2\theta$

$D\eta = 1 - 2\theta - 1 + 4\theta - 4\theta^2 = 2\theta - 4\theta^2 > 0$

$D\tilde{\theta}_2 = \frac{1}{2n} \cdot (\theta - 2\theta^2) \xrightarrow[n \to \infty]{} 0 \Rightarrow$ состоятельно.

6) Проверим модель на регулярность

6.1) $p(x, \theta) = \theta$ – непр. дифф

$p_2(x, \theta) = \frac{1}{2} - \theta$ – непр. дифф.

6.2) $\int_{-1}^{1} \frac{\partial p}{\partial \theta} dx + \frac{\partial p_2}{\partial \theta} \cdot 2 = \int_{-1}^{1} dx + 2 \cdot (-1) = 0$

6.3) $I(\theta) = M\left(\frac{\partial \ln p(x, \theta)}{\partial \theta}\right)^2 =$

$= \int_{-1}^{1} \left(\frac{1}{\theta^2}\right) \cdot \theta \, dx + 2 \cdot \left(\frac{-1}{\frac{1}{2} - \theta}\right)^2 \cdot \left(\frac{1}{2} - \theta\right) = \frac{2}{\theta} + \frac{2}{\frac{1}{2} - \theta} =$

$= \frac{1 - 2\theta + 2\theta}{\left(\frac{1}{2} - \theta\right)\theta} = \frac{2}{(1 - 2\theta)\theta} > 0$ и непрер. на $(0, \frac{1}{2})$ $\Rightarrow$

$\Rightarrow$ регулярна.

7) Проверим оценки на регул.

7.1) обе оценки несмещ.

7.2) модель регул.

7.3) $D\tilde{\theta}_1 = \frac{1+\frac{2}{3}\theta-4\theta^2}{4n}$ - огранич. на V компакте из Θ

$D\tilde{\theta}_2 = \frac{\theta-2\theta^2}{2n}$ — // —

$\Rightarrow$ обе оценки регул.

8) можем применить рав-во Крамера-Рао:

$$D\tilde{\theta}_1 = \frac{1+\frac{2}{3}\theta-4\theta^2}{4n} > \frac{(\theta')^2}{nI(\theta)} = \frac{(1-2\theta)\theta}{2n}$$

$$1+\frac{2}{3}\theta-4\theta^2-2\theta+4\theta^2 = 1-\frac{4}{3}\theta > 0$$

$D\tilde{\theta}_2 = \frac{\theta-2\theta^2}{2n} = \frac{\theta-2\theta^2}{2n}$ — эффективна